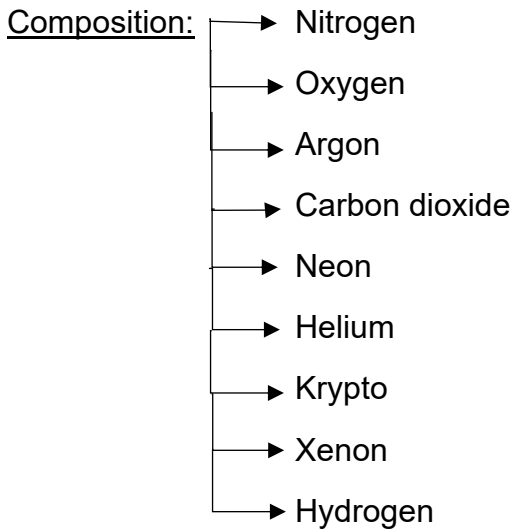


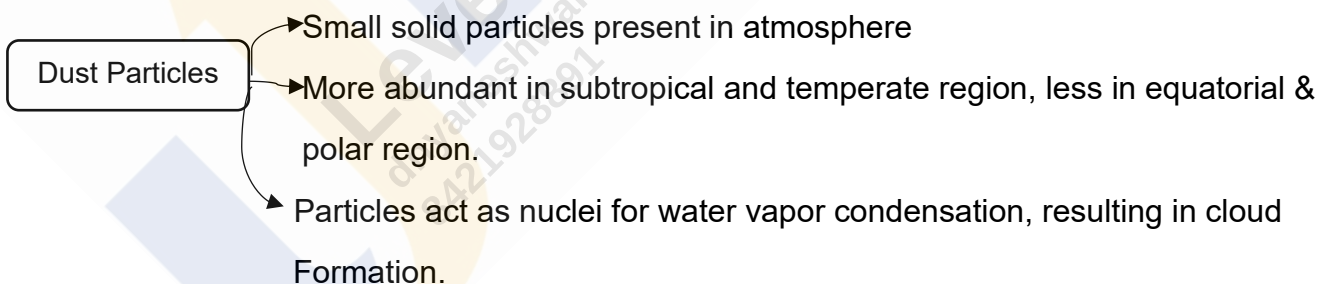
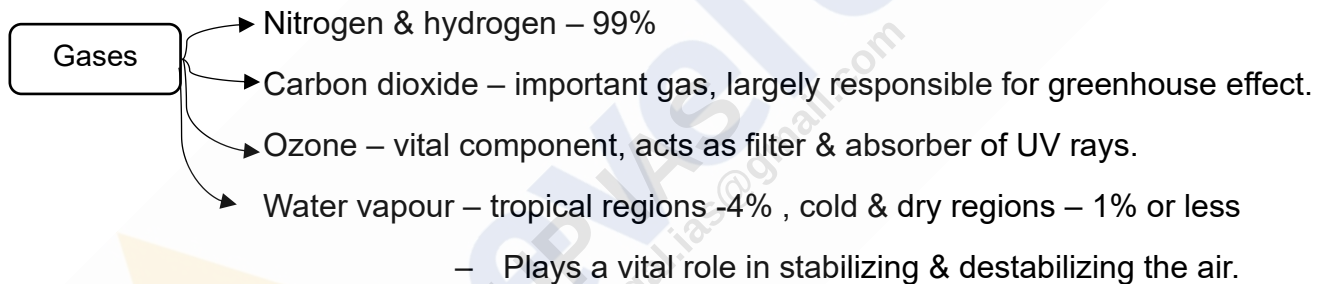
Composition of Earth's atmosphere

Atmosphere is an important part of the earth & composed of gases, water vapour & aerosols.

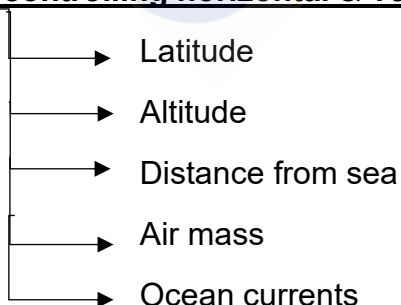


→ Minor gases – CO₂, H₂ and O₃ --- are controlled by reactions in stratosphere (by UV radiation)

→ Atmosphere plays a vital role in supporting life on earth.



Factors controlling horizontal & vertical distribution of temperature



Distribution of temperature

Isotherms: lines connecting areas with the same temperature.

- January → isotherms deviate north over oceans and south over continents
- July → isotherms follow latitude lines more closely

Insolation: Incoming solar radiation, short wave radiation

Heat equator: Also called as thermal equator.

It circumscribes the earth & connects all points of high mean annual temperature

Factors influencing variations of insolation:

- Rotation of the earth
- Angle of inclination
- Length of day
- Land configuration
- Transparency of land

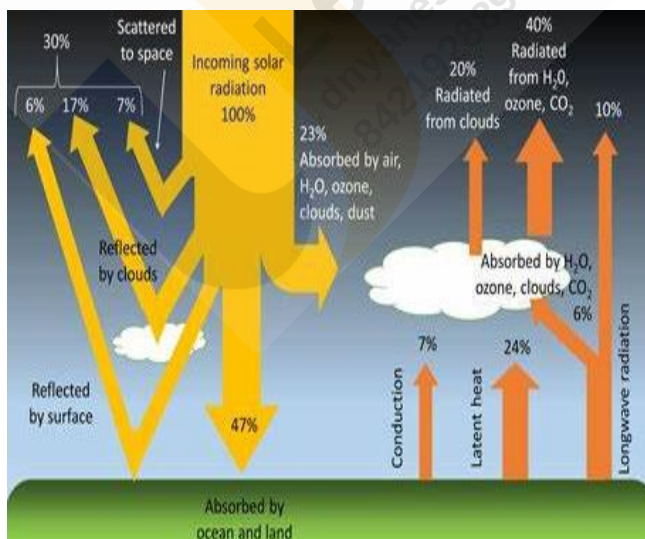
Atmosphere heating & cooling

- (i) Radiation
- (ii) Conduction
- (iii) Convection
- (iv) Advection

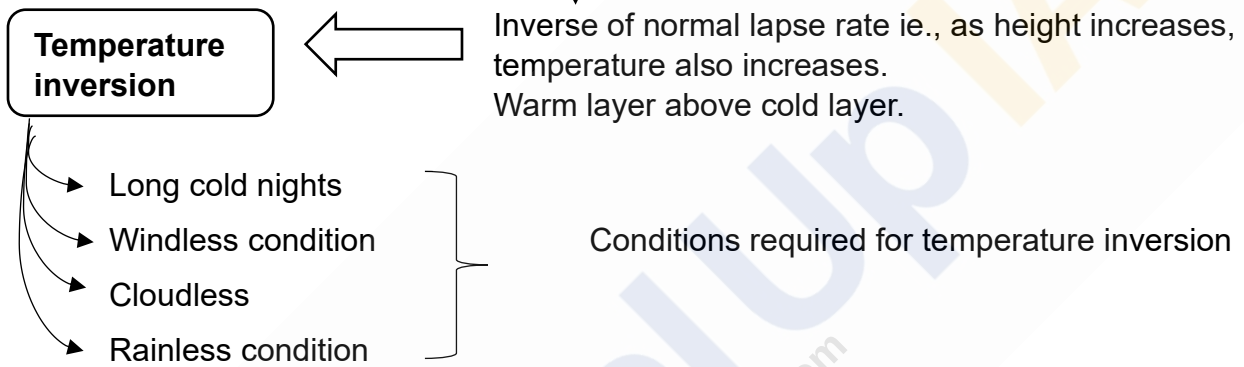
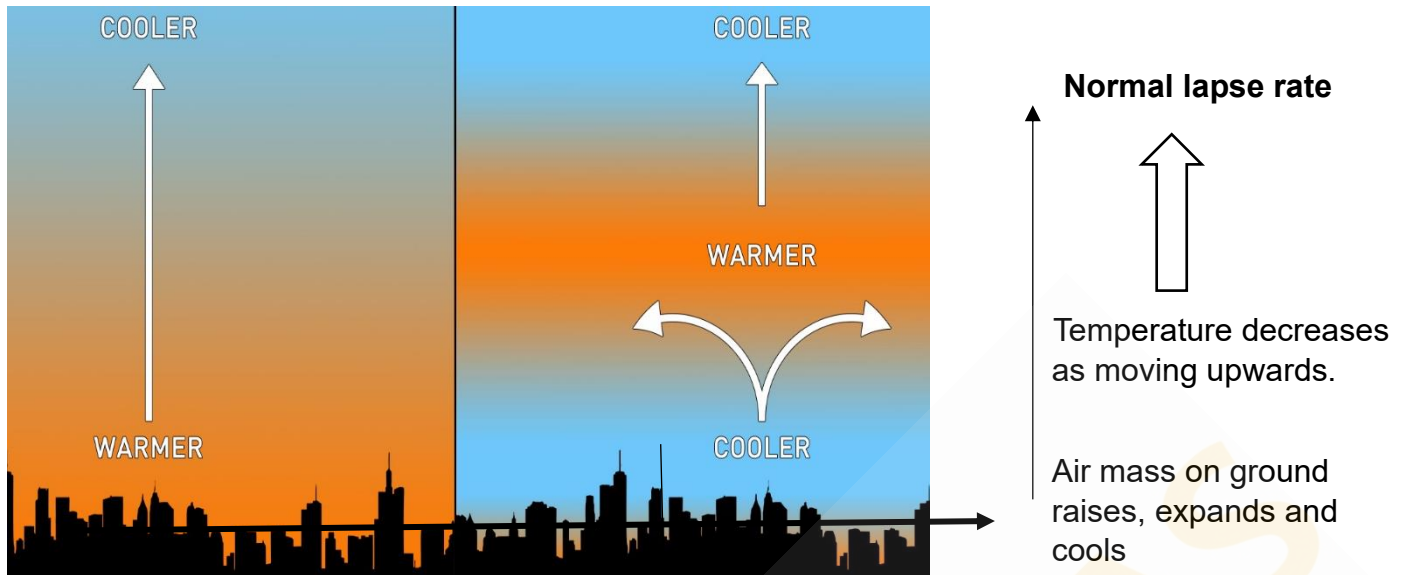
Terrestrial radiation

- Insolation is radiated back in the form of long wave.
- Most of this radiation is absorbed by ozone → heating of atmosphere and sustains life of earth.

Heat budget: Explained in class



Temperature Inversion



Types of temperature inversion:

- Radiation inversion
- Atmospheric inversion
- Frontal inversion
- Valley inversion
- Surface inversion

Impacts:

- Atmospheric stability
- Aridity
- Fog
- Chemical and mechanical geomorphic features
- Agricultural activities
- Navigation
- Smog
- Health

Atmospheric circulations:

- Movement of air within earth's atmosphere.
- Movement is driven by temperature and pressure.
- Circular patterns : Hadley cell, ferrel cell, polar cell, walker cell.
- Jet streams are also part of this.
- Responsible for weather patterns.

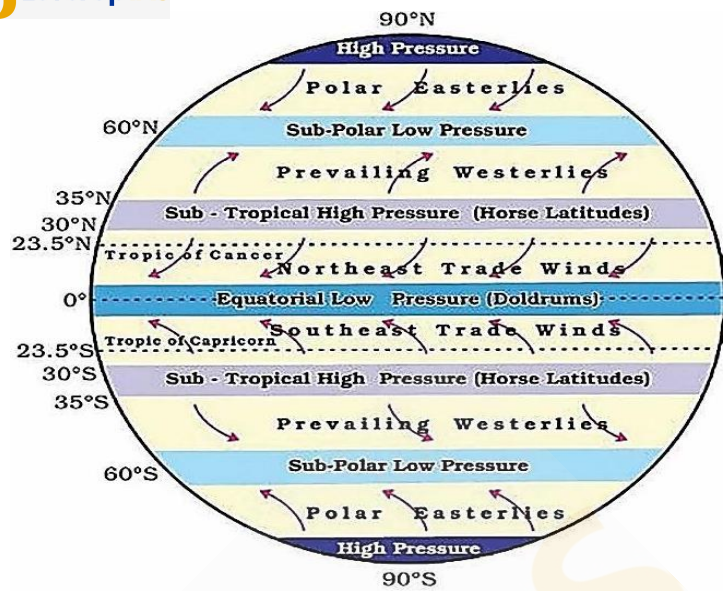
Types of atmospheric circulations:

1. Primary circulations
2. Secondary circulations
3. Tertiary circulations

Wind system

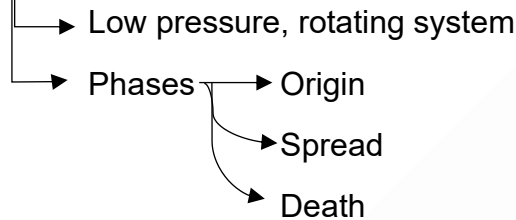
shifting of pressure belts

Jet streams → IN CLASS

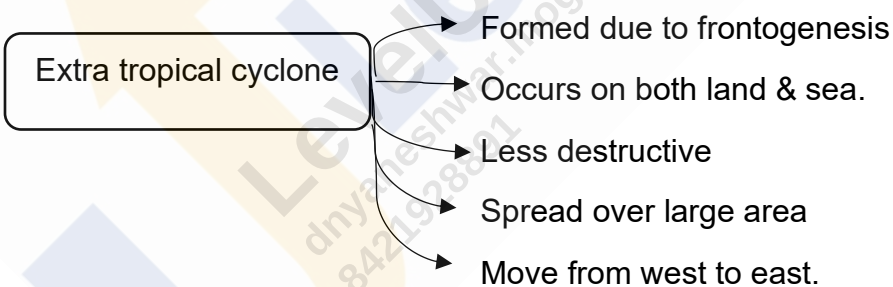
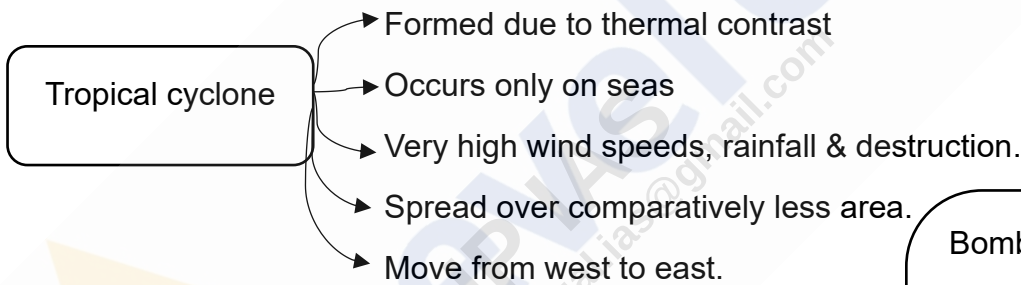


Major Pressure Belts and Wind System

Cyclones



Types of cyclones: (i) Tropical cyclone (ii) Extra tropical cyclones



Bomb cyclone

Reasons for high frequency of cyclone in BOB

Recurving cyclone

Climate change

Air masses → Explained in class

Pre-requisites for cyclone formation

- Low pressure
- Sea surface temperature > 27°C
- Coriolis force
- Moisture source

Coriolis Force	General	Cyclones	Oceanic Currents
Northern hemisphere	Right	Anti clock wise	Clock wise
Southern hemisphere	Left	Clock wise	Anti clock wise